

LR-Newaz Version 1.0

Complete Cinematic Tutorial Storyboard (14 Scenes)

Scene 01: Cinematic Introduction & Brand Identity

Timecode	00:00 - 00:35 (35s)	Transition	Smooth cinematic
	* Screen Action: High-contrast dynamic launch animation of LR-NewazTpoybf Version 1.0. The screen zooms in smoothly onto the Bangladesh Land Record Extractor emblem, transitioning into the application's clean Material 3 Parchment & Sapphire theme. * Camera Movement: Smooth slow zoom-in from 100% to 115%, soft aperture depth-of-field, subtle floating light particles on dark sapphire background. * On-Screen Text: LR-NewazTpoybf v1.0 Bangladesh Land Record Intelligence Platform * Narration (EN): Welcome to LR-Newaz Version 1.0, the definitive forensic intelligence platform designed specifically for Bangladesh land records. From historical handwritten Khatians to complex Cadastral survey maps, this complete user tutorial will guide you step-by-step through every feature. * UI Components: Splash Logo, App Title, Offline Verification Badge, Build Version 1.0.0.		

Scene 02: Standalone APK Installation & Zero-Permission Privacy

Timecode	00:35 - 01:15 (40s)	Transition	Clean lateral s
	* Screen Action: User downloads app-debug.apk from the portal, opens Android Package Installer, confirms installation with zero unnecessary permissions, and launches the app directly into offline mode. * Camera Movement: Top-down 45-degree angle macro zoom focusing on the Android Package Installer modal, panning down to the 'Install' and 'Open' buttons. * On-Screen Text: Step 1: Install Standalone APK (app-debug.apk) 100% Zero-Permission Privacy * Narration (EN): Getting started is completely seamless. Download the standalone APK file. LR-Newaz requires zero dangerous media permissions, strictly utilizing Android's scoped Storage Access Framework to keep all your family deed records 100% private on your own device. * UI Components: Package Installer, App Icon, 'Install' Button, Android Home Screen Launcher Icon.		

Scene 03: Language Selection & Tesseract Offline OCR Engine Setup

Timecode	01:15 - 02:00 (45s)	Transition	Subtle zoom-out
	* Screen Action: User opens the top AppBar menu, taps the MenuBook icon (testTag: 'tesseract_setup_button'), navigates to TesseractSetupScreen, verifies Bengali (ben.traineddata) and English (eng.traineddata) language packs, and checks the green verified engine status indicator. * Camera Movement: Close-up macro pan from top right AppBar icon down to the language pack verification cards with soft glow effects on the status badge. * On-Screen Text: Step 2: Language & OCR Pack Setup Tesseract 5.x Offline Traineddata * Narration (EN): Upon first launch, tap the OCR book icon in the top bar to inspect your offline language packs. The pre-bundled Bengali and English trained data files guarantee lightning-fast character recognition without needing an active internet connection. * UI Components: TopAppBar 'tesseract_setup_button', TesseractSetupScreen, Bengali Pack Status, Engine Initialized Card.		

Scene 04: Home Dashboard, Forensic Metrics & Legal Disclaimer

Timecode	02:00 - 02:40 (40s)	Transition	Split-second cr
* Screen Action: Overview of HomeScreen. Demonstration of the mandatory Legal Disclaimer Banner, real-time forensic metric counters (Total Documents, Verified, Zero-Byte Errors), and the Quick CSV External Reporting bar. * Camera Movement: Gentle downward vertical pan across the Legal Disclaimer, metric counter cards, and search bar. * On-Screen Text: Step 3: Home Dashboard Overview Real-Time Audit Metrics * Narration (EN): The Home Dashboard provides an instant snapshot of your archive. Notice the legal disclaimer highlighting automated forensic assistance, alongside real-time metric cards showing total documents, verified status, and zero-byte error detection. * UI Components: LegalDisclaimerBanner, MetricCard (' ', ' ', ' '), 'btn_export_csv_quick'.			

Scene 05: Bulk Document Ingestion & Storage Access Framework (SAF)

Timecode	02:40 - 03:30 (50s)	Transition	Card expansion
* Screen Action: User taps the Extended FAB ' (SAF)' (testTag: 'fab_saf_folder') or the Bulk Upload icon (testTag: 'btn_bulk_upload_nav'). User selects a batch folder containing PDF deeds and JPEG scans. The upload queue displays live progress, integrity hashing, and zero-byte safety validation. * Camera Movement: Dynamic push-in on the Floating Action Button, transitioning to the SAF system tree, then panning down the active ingestion queue. * On-Screen Text: Step 4: Bulk Ingestion via SAF Scoped Storage & Zero-Byte Safety * Narration (EN): To import records, tap the 'Input Folder' floating button. Select an entire folder of scans or PDF deeds. The bulk upload engine validates each file, checks cryptographic SHA-256 hashes, and filters out corrupted or zero-byte files automatically. * UI Components: FAB 'fab_saf_folder', BulkUploadScreen, ProgressBar, 'btn_start_batch_processing'.			

Scene 06: High-Contrast Document OCR & Adaptive Binarization

Timecode	03:30 - 04:20 (50s)	Transition	Vertical scanni
* Screen Action: Demonstration of OCR processing. The app applies automated deskewing, Otsu adaptive binarization to faded ink, and extracts tabular Khatian structures including Mouza, JL Number, CS/SA/RS/BRS classification, and owner shares with confidence scores. * Camera Movement: Cinematic vertical split-screen comparison: Left side showing the faded yellowed historical deed scan, Right side showing the crisp binarized text with green confidence bounding boxes. * On-Screen Text: Step 5: Tabular OCR Extraction Deskewing, Otsu Binarization & Confidence Scoring * Narration (EN): Watch the AI engine handle aged, faded parchment. Through adaptive binarization and morphological filtering, the system extracts tabular Khatian fields, owner names, and share ratios, color-coding high confidence in green and uncertain fields in amber. * UI Components: DocumentDetailScreen, FieldConfidenceChip (VERIFIED / PROBABLE), Extracted Table Grid.			

Scene 07: Module 04: Bangla Handwriting Recognition (HCR) & Marginalia Review

Timecode	04:20 - 05:15 (55s)	Transition	Horizontal slid
* Screen Action: User taps the HCR banner on Home (testTag: 'btn_open_hcr') or navigates to HcrCorrectionScreen. The UI presents cursive Bengali scribbles, official revenue stamps, and marginal endorsements. User reviews confidence chips, edits uncertain tokens, and tags marginal notes.			

- * Camera Movement: Tight macro zoom on cursive Bengali handwriting sample, highlighting token chips with animated tooltip popups showing confidence percentages.
- * On-Screen Text: Step 6: Bangla Handwriting Recognition (HCR) | Marginalia & Revenue Stamp Recovery
- * Narration (EN): Module 04 is our breakthrough Bangla Handwriting Recognition engine. It deciphers cursive scribbles, revenue stamps, and marginal endorsements. You can inspect individual token confidence scores, make real-time corrections, and insert forensic provenance tags.
- * UI Components: HcrCorrectionScreen, 'card_hcr_banner', TokenConfidenceChip, MarginaliaCard, 'btn_save_hcr_corrections'.

Scene 08: Module 05: Cadastral Survey Map Intelligence & Polygon Detection

Timecode	05:15 - 06:10 (55s)	Transition	Holographic gri
* Screen Action: Demonstration of Cadastral Map GIS detection. The engine analyzes CS, SA, RS, and BS survey sheets, detecting plot polygons, Bengali digit plot numbers, road corridors, water bodies, and the official north arrow with sub-pixel contour approximation. * Camera Movement: Smooth 2D pan across a high-resolution Cadastral survey map sheet, zooming into Dag plot boundaries as glowing vector outlines snap into place. * On-Screen Text: Step 7: Cadastral Map Intelligence Vector Polygon Extraction & Bengali Digit Detection * Narration (EN): In Module 05, the Cadastral Survey Map Intelligence engine processes complex survey sheets. It automatically vectors plot contours, recognizes Bengali Dag numbers, identifies water bodies and roads, and aligns the true north orientation. * UI Components: CadastralMapView, DagPolygonHighlight, RoadNetworkOverlay, NorthArrowIndicator.			

Scene 09: Module 06: Warishnama Inheritance Matching & Genealogical Links

Timecode	06:10 - 07:15 (65s)	Transition	Accordion fold
* Screen Action: User taps ' ' (testTag: 'btn_open_warish_matcher') to enter WarishMatcherScreen. User selects a Warish certificate, cross-references genealogical heirs against CS/SA/RS ancestor titles, reviews legal share calculations (---), and inspects forensic audit tiers. * Camera Movement: Medium pan following the genealogical tree hierarchy descending from deceased landholder to lawful heirs with connecting gold links. * On-Screen Text: Step 8: Warishnama Inheritance Matching Muslim & Hindu Succession & Forensic Evidence Tiers * Narration (EN): Module 06 delivers automated Warishnama inheritance matching. It calculates precise shares under statutory inheritance rules, cross-matches genealogical heirs against historical Khatians, and classifies links into forensic tiers from Verified to Record Gap. * UI Components: WarishMatcherScreen, 'btn_open_warish_matcher', HeirShareCard, EvidenceTierBadge (VERIFIED / CORROBORATED / GAP).			

Scene 10: Cadastral Dag Highlighting & Interactive Parcel Inspection

Timecode	07:15 - 08:05 (50s)	Transition	Circular iris z
* Screen Action: Within WarishMatcherScreen and AtlasTimelineScreen, user taps on an heir's entitlement. Instantly, corresponding Dag plots (e.g. Dag 402, 405) illuminate in bright emerald green on the Cadastral map canvas, while adjacent plots pulse in amber. * Camera Movement: Dynamic 3D tilt and rapid smooth zoom focusing on Dag Plot #402 as the emerald highlight pulses with a golden perimeter halo. * On-Screen Text: Step 9: Highlight Matching Lands Emerald Green Highlighting of Validated			

Dag Plots

- * Narration (EN): Notice how the system immediately bridges records and geography. Tapping an heir's entitlement instantly highlights their validated land plots on the Cadastral map in vivid emerald green, allowing surveyors and landowners to inspect exact physical boundaries.
- * UI Components: InteractiveMapCanvas, HighlightedDagPlot (' '), LandParcelPopup, AreaSummaryBadge.

Scene 11: Module 13 & 14: Historical Property Atlas & Forensic Validation Engine

Timecode	08:05 - 09:05 (60s)	Transition	Time-lapse slid
* Screen Action: User opens Historical Atlas from Home (testTag: 'card_atlas_banner'). Demonstrates the 4-era timeline (CS 1910 -> SA 1956 -> RS 1972 -> BS 1998), visualizing how Dag 402 was partitioned into sub-plots 402/1 and 402/2. Shows discrepancy alerts for area mismatch and honorific name variations. * Camera Movement: Horizontal tracking camera sliding along the timeline chronometer, followed by split-screen tree branching visualization. * On-Screen Text: Step 10: Historical Property Atlas CS -> SA -> RS -> BS Chain & Dag Split Detection * Narration (EN): Modules 13 and 14 introduce the Historical Property Atlas and Forensic Validation Engine. Trace your property through every major survey across a century: CS, SA, RS, and BS. The engine automatically detects Dag splits, area discrepancies, and historical name variants. * UI Components: AtlasTimelineScreen, 'card_atlas_banner', TimelineNode (CS/SA/RS/BS), DagSplitFlowDiagram, DiscrepancyAlertCard.			

Scene 12: High-Speed Offline Search & Multi-Criteria Filtering

Timecode	09:05 - 09:50 (45s)	Transition	Snappy dissolve
* Screen Action: User returns to HomeScreen, types ' ' or ' ' into the search field (testTag: 'search_input'). Instant full-text search results filter the document list within milliseconds using local SQLite FTS. * Camera Movement: Close-up on typing in the search bar, with instant reactive filtering of the list below, accompanied by soft amber highlight pulses on matched keywords. * On-Screen Text: Step 11: Instant Offline Search SQLite Full-Text Search (FTS) by Khatian, Dag & Owner * Narration (EN): Finding any deed or plot in your archive is instantaneous. Use the search bar to query by Khatian number, Dag plot, Mouza, or owner name in English or Bengali. SQLite Full-Text Search delivers results locally in under twenty milliseconds. * UI Components: OutlinedTextField 'search_input', CapturedRecordCard, HighlightedSearchTerms.			

Scene 13: Multi-Format Forensic Report Export (PDF, Excel, GeoJSON, ZIP)

Timecode	09:50 - 10:45 (55s)	Transition	Smooth 3D page
* Screen Action: User taps ' (SAF)' or opens DocumentDetailScreen export menu. User selects PDF Court Dossier, Excel XLSX Workbook, Cadastral GeoJSON, or All-in-One Forensic ZIP Archive. Android SAF dialog saves the files to device storage with zero permissions. * Camera Movement: Hero showcase view of the generated multi-page bilingual PDF dossier, zooming into the forensic stamp, followed by Excel and GeoJSON asset cards. * On-Screen Text: Step 12: Multi-Format Forensic Export Court Dossier PDF, Excel, GeoJSON & ZIP * Narration (EN): Exporting court-ready reports is effortless. Choose from professional bilingual PDF Dossiers with cryptographic stamps, comprehensive Excel workbooks, GIS GeoJSON layers, or packaged forensic ZIP archives for legal and surveying submissions.			

* UI Components: ReportExportDialog, 'export_csv_button', FormatRadioGroup (PDF / EXCEL / GEOJSON / ZIP), SuccessSnackBar.

Scene 14: Offline Security, Summary & Next Steps

Timecode	10:45 - 11:30 (45s)	Transition	Slow cinematic
	* Screen Action: Final review of system architecture: 100% offline capability, local Room database, zero cloud leaks, and verified build stability. Shows the official GitHub repository, documentation links, and conclusion. * Camera Movement: Smooth pull-back from mobile device to high-tech desk environment, fading up the LR-NewazTpoybf Version 1.0 golden badge and documentation links. * On-Screen Text: LR-NewazTpoybf Version 1.0 Offline * Private * Forensic * Complete * Narration (EN): You are now fully equipped to utilize LR-Newaz Version 1.0. Completely offline, forensically verifiable, and built to safeguard Bangladesh land heritage. Download the release, explore the documentation, and master your land records with total confidence. * UI Components: FinalSummaryCard, DocumentationLinks, Version1Badge, ExitCredits.		